

4. Potassium chloride and potassium sulfate are salts that can be made from the reactions of acids. Becky and David carried out experiments to prepare both salts.

- (a) To make a sample of potassium chloride, they reacted potassium carbonate powder with dilute hydrochloric acid, HCl. They used the following method.

Stage 1 – add potassium carbonate powder to dilute hydrochloric acid

Stage 2 – filter the mixture

Stage 3 – leave the remaining mixture in a warm place overnight

- (i) David said it was important to know the volume of acid in stage 1.

Circle whether you agree or disagree with his statement and explain your answer. [2]

Agree / Disagree

Explanation

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- (ii) Give the balanced **symbol** equation for the reaction taking place in stage 1. [3]

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- (b) To make a sample of potassium sulfate, they reacted potassium hydroxide solution with dilute sulfuric acid, H_2SO_4 . They used the following method.

Stage 1 – measure out 25.0 cm^3 of potassium hydroxide solution into a flask

Stage 2 – add 4-5 drops of indicator

Stage 3 – record the volume of sulfuric acid required to neutralise the solution

Stage 4 – repeat without the indicator, using the volume of acid used in stage 3

Stage 5 – leave the remaining mixture in a warm place overnight

- (i) Becky was concerned that carrying out stages 1-3 only once would mean that the volume of sulfuric acid used in stage 4 would be larger than that actually needed to neutralise the potassium hydroxide solution. Suggest the reason for her concern.

[1]

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- (ii) Write the **ionic** equation to show how water is formed during this reaction and state the source of each of the ions.

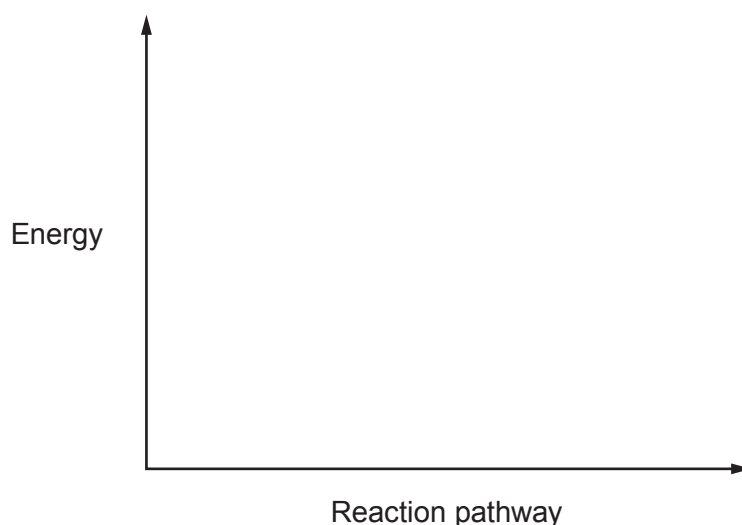
[2]

Equation

Source of ions

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- (iii) The reaction is exothermic. Sketch the energy profile diagram for this reaction. [1]



(c) Becky and David carried out a series of tests to identify the ions present in their salts.

- (i) They carried out a flame test on each of the salts. Give the reason why the results of this test could not be used to identify the salts. [1]

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- (ii) Describe a test that enabled them to tell the salts apart. Give the results seen with both salts. [2]

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